

Career Pathways Platform

Build Document — Prototype & Production Roadmap

Project: Multi-industry career pathways map with AI advisor and live job ingestion pipeline

v1 industries: Additive Manufacturing (lead), Semiconductors · **Post-prototype:** Space

Document version: 2.1 · **Prepared for:** Freelance / contracted developer

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1. What we're building (in plain words)

Two systems that work together.

System A — The Career Pathways Website. A public website where someone (a student, a career changer, a workforce advisor, a policy maker) can pick an industry, see all the jobs in it laid out as a map, click any job to see the career paths leading in and out, view rich detail on each role (skills, salary, education needed), and chat with an AI advisor that recommends pathways based on their background. v1 ships with Additive Manufacturing and Semiconductors. Space is a planned third industry, added after v1 ships.

System B — The Data Ingestion Pipeline. A behind-the-scenes automation that pulls real job postings from company career pages every week, uses AI to read each posting and extract structured information, then validates whether each role fits the existing taxonomy or represents something new. High-confidence matches enrich the website automatically (showing "23 open jobs at 8 companies" on each role). Ambiguous cases route to a human reviewer. This keeps the taxonomy alive, current, and defensible — and makes adding new industries (like Space) 10x cheaper than manual curation.

Both systems run on free-tier infrastructure for the prototype.

Why Additive Manufacturing first: workforce demand is growing fast (America Makes, DoD investment, defense supply-chain reshoring), the talent pipeline is acutely under-mapped, and the role landscape has clean operator → technician → engineer → R&D progression that maps beautifully to the value-chain × seniority grid. Semiconductors second because the taxonomy is already drafted. Space is structurally similar to AM but with stricter regulatory layers — easier to add once the platform is proven.

2. Reference websites — study these

Direct competitors / inspiration

1. DOE Critical Minerals & Materials Career Map URL: <https://criticalmaterials.careerpathplatform.com/map/>

- The closest reference site for the exact UX we're building. Built by Julius Education for the U.S. Department of Energy.

- Study: the value-chain x seniority grid layout, role detail panels, degree-required signaling, click-chain pathway visualization.
- This is the visual quality bar for our v1.

2. E3 Semiconductors Career Map (Julius Education) URL: <https://e3semi.careerpathplatform.com/map/>

- Same platform as #1, instantiated for the semiconductor industry. This is our direct competitor for the semi vertical.
- Study: how Julius adapts the same grid framework for a different industry — clusters, role names, pathway choices. Compare side-by-side with #1 to see how the architecture stays constant while the content varies.
- Architectural takeaway: Julius runs a multi-tenant platform where each industry is a subdomain (`{industry}.careerpathplatform.com`). This validates our multi-industry architecture. Plan to differentiate on: AI advisor, live job ingestion, transparency of transition logic.

3. mySkills myFuture (US Dept of Labor) URL: <https://www.myskillsmyfuture.org/>

- Government-grade career transition tool. Type your current job, see related careers and skill gaps.
- Study: the "transferable skills" framing, occupation-to-occupation transitions.

4. CareerOneStop URL: <https://www.careeronestop.org/>

- US Department of Labor's flagship career exploration site. Massive data, dated UI.
- Study: how they integrate BLS labor market data (wages, projections) and link to training.

5. O*NET OnLine URL: <https://www.onetonline.org/>

- The canonical US occupational database. Every role classified by skills, knowledge, work activities.
- Study: the data model itself — this is what serious career platforms reference for credibility. Search "additive manufacturing" and "semiconductor" here for baseline SOC codes.

Additive Manufacturing — industry references

6. America Makes URL: <https://www.americamakes.us/>

- The national accelerator for AM. Workforce reports, training partner directory, member companies.
- Study: their "Workforce & Education" content for canonical role names, certifications, and the AM Body of Knowledge.

7. ASTM Additive Manufacturing Center of Excellence URL: <https://amcoe.org/>

- Standards body for AM. Hosts the AM-CF (Additive Manufacturing Certificate Framework) — the closest thing to canonical AM job/skill definitions.
- Study: certifications by role family. These map directly to our skill ontology.

8. AMUG (Additive Manufacturing Users Group) URL: <https://www.amug.com/>

- Practitioner community. Conference proceedings show what real AM teams look like at GE, Boeing, Lockheed, Stryker, etc.

9. Wohlers Report (industry analyst) URL: <https://wohlersassociates.com/>

- The industry-standard annual report on AM. Pricey, but excerpts are public and credible to cite in stakeholder reports.

Semiconductor industry references

10. SEMI workforce resources URL: <https://www.semi.org/en/workforce-development>

- The semiconductor industry association's workforce hub. Job role definitions, certifications, training partners.

11. CHIPS Act workforce reports (SIA) URL: <https://www.semiconductors.org/>

- SIA publishes workforce gap analyses — credible references for the semiconductor taxonomy and salary bands.

Visual / interaction design references

12. Information is Beautiful — Career Maps URL: <https://informationisbeautiful.net/>

- For learning what compelling visual storytelling around career data looks like.

13. Sankey-style career flow examples URL: <https://flourish.studio/examples/sankey-diagram/>

- Reference for showing flow between roles when we add the "where do people go from here" feature.

14. Climatebase URL: <https://climatebase.org/>

- Industry-specific career platform (climate). Excellent example of a niche vertical done well: jobs board, learning paths, community.
- Study: how they balance content depth with clean UX, and how they monetize.

15. Built In URL: <https://builtin.com/jobs>

- Tech-industry-specific careers + content site. Strong SEO, high traffic.
- Study: their company profiles and salary tools.

AI agent / chat UX references

16. Perplexity URL: <https://www.perplexity.ai/>

- Reference for AI answers with cited sources. Our agent should cite which roles/pathways it's drawing from.

17. Claude.ai (this product) URL: <https://claude.ai/>

- Reference for clean conversational interface, streaming responses, message structure.

Live job ingestion references

18. Greenhouse public job board API URL: <https://developers.greenhouse.io/job-board.html>

- Free public API. Used by many AM and semi companies (Velo3D, Carbon, Desktop Metal, Intel, Micron, etc.). Single source for thousands of roles.

19. Lever public postings API URL: <https://github.com/lever/postings-api>

- Same idea, smaller and mid-stage companies — common for AM startups.

20. SerpAPI (paid, for production scale) URL: <https://serpapi.com/google-jobs-api>

- Production-grade job aggregation. ~\$50/mo. Worth it when you scale, especially for companies that don't expose public ATS APIs (most large defense primes).

Workforce data sources to integrate

21. BLS Occupational Employment & Wage Statistics (OEWS) URL: <https://www.bls.gov/oes/>

- Free. Authoritative US salary and employment data by occupation and metro area.

22. Lightcast (formerly EMSI Burning Glass) URL: <https://lightcast.io/>

- Production-grade labor market data API. Paid (\$\$\$). Where you go when CHIPS-Act- or DoD-funded clients ask for state-level workforce projections.

Tech stack documentation

23. Next.js 14 docs URL: <https://nextjs.org/docs>

- The web framework we'll use.

24. Anthropic Claude API docs URL: <https://docs.claude.com/>

- The AI model and SDK powering the agent and the ontology matcher.

25. Supabase docs URL: <https://supabase.com/docs>

- Postgres database + auth + vector search. Free tier hosts the prototype.

26. Vercel docs URL: <https://vercel.com/docs>

- Where we deploy. Free tier hosts the prototype.

3. System architecture

USER (browser)



WEB APP (Next.js on Vercel)

Pages:

/	→ industry chooser
/[industry]	→ career map view
/[industry]/role/[id]	→ role detail
/admin	→ review queue (auth-gated)

Components:

<CareerMap>	click-chain pathway grid
<RoleDetailModal>	skills, salary, training, jobs
<AgentChat>	streaming Claude advisor
<ReviewQueue>	admin: approve/reject matches

API routes:

POST /api/agent/chat	Claude API + RAG
GET /api/industries	list
GET /api/industries/:id	roles, pathways, jobs
POST /api/admin/decide	approve/reject from review



DATABASE (Supabase Postgres + pgvector)

Tables:

industries	(id, name, description)
canonical_roles	(id, industry_id, title, cluster, seniority, salary, skills, description, embedding)
pathways	(id, industry_id, name, role_ids[])
raw_jobs	(id, source, company, raw_title, raw_description, url, scraped_at)
extracted_jobs	(id, raw_job_id, normalized_title, skills, seniority, location)
role_matches	(id, extracted_job_id, canonical_role_id, confidence, status: pending/approved/rejected)
review_decisions	(id, match_id, decided_by, decided_at)



INGESTION PIPELINE (GitHub Actions cron, weekly)

Step 1: Scrapers

- Greenhouse API for: Velo3D, Carbon, Desktop Metal, Intel, Micron, GlobalFoundries, AMD, NVIDIA...
- Lever API for: smaller AM companies
- Playwright fallback for outliers (defense primes, legacy ATS)

Output → raw_jobs table

Step 2: Extractor (Claude Haiku for cost)

For each raw_job: structured JSON extraction

Output → extracted_jobs table

Step 3: Ontology Matcher (Claude Sonnet for accuracy)

For each extracted_job:

- Find top-3 canonical_roles by vector similarity
- Claude judges: match? new role? noise?
- Confidence score 0-1

Output → role_matches table

Step 4: Routing

- confidence > 0.85 → auto-approve, status='approved'
- 0.50-0.85 → status='pending', queue for human review
- < 0.50 → status='rejected'

Step 5: Aggregation

Update canonical_roles with current job counts, company list, recent posting samples

4. Prototype build plan (free-tier, ~3 weeks)

Assumes Claude Code installed locally; double calendar time if working chat-only.

Week 1 — Foundation and the website

Day	Goal	Concrete deliverable
1	Setup	GitHub repo, Next.js + TS + Tailwind scaffold, Vercel deploy, Supabase project, Anthropic key, env vars wired. Domain purchased (~\$12/yr) and pointed at Vercel. Empty homepage live at the custom domain with HTTPS.
2	Map component (part 1)	Build <code><CareerMap></code> React component: value-chain × seniority grid, role nodes, SVG pathway layer. Static layout, no interaction yet.
3	Map component (part 2)	Click-chain logic, SVG line drawing, tooltips, dim effect, degree badges. Matches the visual spec quality bar.
4	AM taxonomy + data layer	Build the Additive Manufacturing taxonomy (~50 roles, 5 clusters, 10-12 pathways) using the same JSON shape as Semiconductors. Map renders it.
5	Industry switcher	<code>/additive-manufacturing</code> and <code>/semiconductors</code> routes. Same map, different data. Industry selector in header.

Week 2 — Role detail, agent

Day	Goal	Concrete deliverable
6	Role detail modal	Click any role → modal shows description, salary range, degree, skills, pathways it appears in, adjacent roles. Closes cleanly.
7	Filters and search	Filter by degree level (HS / 2yr / 4yr+), filter by cluster, full-text search across role titles and skills.
8	Mobile + polish pass	Responsive layout, accessibility basics (keyboard nav, focus states, ARIA), meta tags.
9	Agent backend	<code>POST /api/agent/chat</code> endpoint. Claude API call with system prompt. RAG: load all roles+pathways for active industry into context. Streaming response.
10	Agent frontend	Chat sidebar component. Message history, streaming UI, "suggested prompts" buttons, citations linking back to roles.

Week 3 — Ingestion pipeline + polish

Day	Goal	Concrete deliverable
11	Scraper foundation	Python project in same repo (or sibling). Fetch Greenhouse public API for ~10 AM companies (Velo3D, Carbon, Desktop Metal, Markforged, etc.) and ~10 semi companies (Intel, Micron, AMD, NVIDIA, GF, etc.). Store raw to Supabase.
12	Extractor	For each raw job, call Claude Haiku to extract structured fields. Store to <code>extracted_jobs</code> .
13	Ontology matcher	Embed all canonical roles into pgvector. For each extracted job, find top-3 by similarity, ask Claude Sonnet to judge match. Store to <code>role_matches</code> .
14	Review UI	<code>/admin</code> page (password-gated). Pending matches list, approve/reject buttons, view raw posting alongside proposed match.
15	Cron + integration	GitHub Actions workflow runs the pipeline weekly. Website displays "X open roles at Y companies" on each canonical role from approved matches.
16	Polish + deploy	Mobile responsive, meta tags, OG images, README, license, env-vars-example file.
17	Buffer / bug fixes	Always needed.
18	Demo prep	Loom walkthrough, stakeholder report (see template below), client-handover checklist. Space industry documented as next-up scope (research links, candidate company list, expected timeline).

Stakeholder report template (filled in during week 3)

A 5-page PDF for clients/funders covering:

1. **Executive summary** — what was built, live URL, key numbers (industries, roles, pathways, jobs ingested)
2. **Architecture diagram** — the system architecture above, simplified
3. **Live cost dashboard** — current free-tier usage, projected costs at 1k / 10k / 100k users
4. **Demo script** — the 90-second story to tell the website
5. **Roadmap** — Space industry as next deliverable, plus production-readiness path to 10k users

5. Tech stack — prototype vs production

Layer	Prototype (free, \$0-30/mo)	Production (10k+ users, \$500-2k/mo)
Frontend hosting	Vercel Hobby	Vercel Pro (\$20/user/mo) or self-host on Cloudflare Pages
Framework	Next.js 14 + TypeScript + Tailwind	Same — scales fine
Database	Supabase free (500MB, pauses after 7d idle)	Supabase Pro (25/mo) or AWS RDS Postgres (50-200/mo)
Vector search	pgvector on Supabase free	pgvector Pro, or dedicated Pinecone (\$70/mo) at high RAG volume
Auth	None for v1; Supabase Auth when needed	Supabase Auth Pro, or Clerk (\$25+/mo) for richer SSO/orgs
AI agent	Claude Sonnet 4.6 via Anthropic API	Same, with prompt caching (\$) and per-org rate limits
Ingestion compute	GitHub Actions cron	Modal (\$30+/mo), or AWS Lambda + EventBridge
Scraping	Greenhouse + Lever public APIs (free)	Above + SerpAPI (\$50-200/mo) for Google Jobs coverage
Email / transactional	Not in v1	Resend (\$20/mo) for digest emails, password resets
Analytics	Vercel Analytics free	PostHog Cloud (\$0-450/mo depending on volume)
Error tracking	Sentry free (5k events/mo)	Sentry Team (\$26/mo)
Domain	Custom domain (\$12/yr, Day 1)	Same domain — yours forever
CDN / DDoS	Vercel built-in	Cloudflare in front of Vercel for advanced WAF
Backup	Supabase free tier auto-backup (7d retention)	Supabase Pro (point-in-time restore) + S3 cold backups

6. Cost breakdown

Prototype (during build + first 90 days post-launch)

Item	Monthly cost	Notes
Vercel Hobby	\$0	Non-commercial ToS — flag for handover
Supabase Free	\$0	500MB DB, 5GB bandwidth
GitHub	\$0	Public repo, or private with 2k Actions min/mo
Anthropic API — agent	\$5-25	Claude Sonnet, ~500-2,000 chats/mo
Anthropic API — pipeline	\$10-30	Claude Haiku for extraction, Sonnet for matching, ~1k jobs/week
Vercel Analytics	\$0	Free tier
Domain	\$1/mo	\$12/yr, registered Day 1 — yours forever across all infrastructure changes
Total	~\$16-56/mo	Realistic prototype operating cost

At 10,000 monthly users (production)

Assumptions: 10,000 MAU, 30% engage with the agent (3,000 agent users), 5 chats per agent user, pipeline processes 5,000 new jobs per week.

Item	Monthly cost	Notes
Vercel Pro	\$20	1 seat — required for commercial use
Vercel bandwidth overage	\$0-40	1TB included, depends on traffic
Supabase Pro	\$25	8GB DB, point-in-time restore
Supabase compute (if scaling reads heavily)	\$0-60	Most apps stay in base
Anthropic — agent	\$250	15,000 chats × ~\$0.017
Anthropic — pipeline	\$80	20,000 jobs/mo extracted+matched
Anthropic — prompt caching savings	-\$60	Cache the system prompt + taxonomy context
Sentry Team	\$26	Error tracking
PostHog Cloud	\$0-80	Free up to 1M events
Resend	\$20	Transactional email
Cloudflare (optional)	\$0	Free tier sufficient
Domain	\$1	Annualized
Total	~\$420-600/mo	At 10k MAU

Cost levers if budget is tight at scale

- **Prompt caching** on Anthropic — cuts agent costs 50-80% when the system prompt + taxonomy are stable
- **Use Haiku 4.5** for the agent on simple queries, route to Sonnet only for complex ones — 5-10x cheaper
- **Cache common queries** in Redis — career-advisor questions follow patterns
- **Pipeline runs less frequently** (biweekly instead of weekly) on stable industries

7. Production roadmap — scaling to 10,000+ users

Phase 1 — Prototype to "soft launch" (months 1-2)

What changes after the prototype is live:

- **Add the Space industry.** This is the immediate post-prototype deliverable. Build the Space taxonomy (~50-60 roles spanning launch ops, satellite manufacturing, propulsion, mission ops, ground systems), source ~10-15 space companies for the ingestion pipeline (SpaceX, Blue Origin, Rocket Lab, Relativity, Anduril Space, Maxar, Planet, Astranis, plus primes like Lockheed Space and Northrop Space), validate with at least one space-industry SME before launch.
- **Add authentication.** Supabase Auth for users to save pathways, get email digests, track progress.
- **Move Vercel Hobby → Pro.** Required for commercial use. Unlocks 60s function timeouts (helps RAG).
- **Add Sentry.** Catch errors before users report them.
- **Analytics with funnel tracking.** PostHog or similar — measure: which industries get traffic, what users ask the agent, where they drop off.
- **SEO content.** Each role page becomes its own indexed URL (`/additive-manufacturing/role/dmls-machine-operator`). Write 200-400 words of high-quality content per role. ~150 roles × 300 words = 45,000 words. Claude can draft, you edit. This is what drives organic traffic.

Phase 2 — Scaling to 1,000-10,000 users (months 3-6)

Operational changes:

- **Supabase free → Pro** (\$25/mo). Removes the 7-day pause, gives 8GB, point-in-time restore.
- **Add prompt caching** on the agent. Cache the industry taxonomy in the system prompt — first request slower, subsequent ones much cheaper.

- **Implement rate limiting** on the agent endpoint. Prevent runaway bills from abusive users or bots. Use Upstash Redis (free tier) or Supabase row-level rate limit tables.
- **CDN-cache the static taxonomy data.** Roles/pathways change weekly at most — serve from Vercel edge cache, not API on every load.
- **Set up scheduled DB backups** to S3 (Supabase Pro auto-backups + your own copy).
- **Monitoring dashboard.** Vercel Analytics + Supabase metrics + Anthropic usage. Build a single internal page showing: DAU, agent chats/day, jobs ingested this week, pipeline error rate, monthly cost burn.
- **Content review cadence.** Schedule weekly 1-hour review sessions for the ingestion review queue. Don't let it pile up.
- **Legal review for scraping.** Even with public APIs, get a one-page memo from a lawyer confirming the approach. Especially before pursuing govt or DoD-adjacent contracts (relevant for AM defense work and Space).

Phase 3 — 10,000+ users and revenue (months 6-12)

Bigger architectural moves:

- **Multi-tenancy.** Workforce boards, community colleges, and corporate L&D teams want their own branded views. Add an `organizations` table, support custom subdomains (`workforce.partner.com`).
- **Lightcast/Emsi integration.** State-level wage and projection data. Required credibility for selling to government and education. ~\$10-30k/yr API license.
- **Move ingestion to a real queue system.** GitHub Actions cron is fine through ~50,000 jobs/week. Beyond that, move to Modal, Inngest, or a self-hosted worker on Fly.io. Adds retry, backpressure, observability.
- **Database read replicas.** Supabase Pro supports this. Reads dominate traffic; replicas keep p95 latency low.
- **Background job processing for heavy agent calls.** Long-running RAG queries should be enqueued, not run inline. Streaming the answer back via Server-Sent Events.
- **A/B testing infrastructure.** PostHog or LaunchDarkly. Test agent prompts, page layouts, conversion flows.
- **Customer support tooling.** Linear or HubSpot for managing inbound. Even at 10k MAU, expect 5-20 support emails per week.
- **SOC 2 prep** if selling to enterprises or DoD-adjacent buyers. Vanta or Drata, ~\$10k/yr starting. Skip until you have an enterprise pipeline.
- **Accessibility audit.** WCAG 2.1 AA compliance is non-negotiable for government buyers. Hire an audit firm or use Deque axe (\$\$).

Phase 4 — Beyond 10,000 (months 12+)

If/when you're sure this is a business:

- **Move database off Supabase to AWS RDS or Aurora.** Supabase is great until you outgrow it; large customers may demand AWS-region-specific data residency (especially Space/defense customers needing GovCloud).
- **Dedicated AI inference contracts.** At enough scale, Anthropic offers committed-spend discounts (talk to their sales).
- **Build a public API.** Workforce boards and EdTech partners will want programmatic access to taxonomy and pathways. Stripe + API key management.
- **Mobile app.** React Native — the map is naturally touch-friendly.
- **Internationalization.** Spanish first (large workforce intersection in US AM/manufacturing), then EU languages if expanding.
- **ML on top of taxonomy.** Predict career transitions from cohort data, recommend pathways probabilistically rather than rule-based.

8. Risks and known limitations

Taxonomy quality. v1 taxonomies are generated with industry knowledge but not formally SME-validated. Risk: a workforce board reviewer says "this isn't how we describe roles in our state." Mitigation: budget for SME review (1-2 weeks of paid expert time per industry, ~\$3-5k each) before any government bid. For AM specifically, partner with America Makes or an AM-focused community college program for review.

Scraping legal exposure. Even with public APIs, terms of service can change. Mitigation: log every source URL with timestamp, use only sources with clear public-API status (Greenhouse, Lever, Workday public boards), get legal sign-off before scaling. Defense primes (relevant for AM and Space) often don't expose public ATS — plan for Playwright fallbacks with conservative rate limits.

LLM hallucination in the ontology matcher. Claude is good but not infallible. Risk: a "Process Engineer" job at Tesla gets matched to an AM process engineer role — same words, different industry. Mitigation: industry-scoped matching (a job from a known AM company can only match AM roles), human review threshold tuned conservatively, audit logs on every auto-approval.

Vercel Hobby ToS during prototype. Technically non-commercial. Risk: Vercel sends a "please upgrade" email if they detect commercial use. Mitigation: budget the \$20/mo upgrade and do it before any paid pilot.

Anthropic API outages. Rare but they happen. Risk: agent down for hours. Mitigation: graceful degradation — if API errors, show "Agent temporarily unavailable, here's a static FAQ" instead of broken UI.

Single-person dependency. If you're solo, you're a bus factor of one. Risk: project stalls if you're sick or busy. Mitigation: write down everything (this document is a start), keep all credentials in a shared password manager, set up co-maintainer access on GitHub before you're stretched.

Taxonomy drift between industries. As Space is added in Phase 1, vocabulary inconsistencies creep in. Risk: "QA Technician" means different things in AM vs. semi vs. Space. Mitigation: maintain a cross-industry skill ontology that all industries map into; review quarterly.

Cost surprise from runaway agent. A single bad actor or bot loop can burn through \$100s of API in hours. Mitigation: per-IP and per-user rate limits, daily spend caps in the Anthropic dashboard, alerting at 50% of monthly budget.

AM industry data sparsity vs. semiconductors. AM has fewer large public-ATS companies than semi — many AM jobs sit at smaller startups using less-scrape-friendly tools, or at huge defense primes with locked-down ATS. Risk: thinner live job data on AM roles than semi roles in v1. Mitigation: supplement with curated company list and direct outreach to America Makes member companies for posting access.

9. What "done" means at each stage

Prototype done = all of these are true:

- ☐ Live at a public URL with HTTPS
- ☐ Additive Manufacturing and Semiconductors fully populated
- ☐ Click-chain map UX matches the visual quality bar
- ☐ Role detail modal shows: title, salary, degree, skills, description, pathway membership, adjacent roles
- ☐ AI agent answers career questions with citations to roles
- ☐ Filters (degree, cluster) and search work
- ☐ Mobile responsive
- ☐ GitHub repo has README, env-vars-example, license, deploy instructions
- ☐ Ingestion pipeline runs weekly and produces approved-job counts on at least one industry
- ☐ Admin review queue accessible and functional
- ☐ Stakeholder PDF report delivered
- ☐ Space industry scoped as next deliverable (taxonomy outline + company shortlist documented, not yet built)

Soft launch done = above plus:

- ☐ Space industry live
- ☐ Custom domain
- ☐ Authentication for saved pathways
- ☐ SEO content on every role page
- ☐ Sentry error tracking active
- ☐ Analytics tracking with goals defined

10,000-user production done = above plus:

- ☐ Multi-tenancy support
- ☐ State-level labor data integrated (Lightcast or BLS OEWS)
- ☐ Rate limiting on agent
- ☐ Backup strategy beyond default
- ☐ WCAG 2.1 AA accessibility audit passed
- ☐ Monitoring dashboard with alerting
- ☐ Documented incident response procedure
- ☐ Two people with admin/deploy access

Appendix — Quick reference for handover

Repo layout (target):

```
career-platform/
├─ README.md
├─ LICENSE
├─ .env.example
├─ package.json
├─ apps/
│   └─ web/                                # Next.js website
│       └─ src/
│           ├── app/
│           ├── components/
│           ├── data/                      # taxonomies (JSON)
│           │   ├── additive-manufacturing.json
│           │   ├── semiconductors.json
│           │   └─ space.json             # added post-prototype
│           └─ lib/
│               └─ public/
├─ pipeline/                              # Python ingestion
│   ├── scrapers/
│   ├── extractor.py
│   ├── matcher.py
│   └─ requirements.txt
├─ packages/
│   └─ shared/                            # shared TS types
├─ .github/
│   └─ workflows/
│       └─ ingest.yml                     # weekly cron
```

Required accounts (all free to start, except domain ~\$12/yr):

1. GitHub (github.com)
2. Vercel (vercel.com — sign in with GitHub)
3. Supabase (supabase.com)
4. Anthropic (console.anthropic.com — generate API key, copy once)
5. Domain registrar — Cloudflare Registrar or Porkbun. Buy the domain on Day 1. Stays yours forever across all future hosting.

Required environment variables:

```
ANTHROPIC_API_KEY=sk-ant-...
SUPABASE_URL=https://xxx.supabase.co
SUPABASE_ANON_KEY=eyJ...
SUPABASE_SERVICE_ROLE_KEY=eyJ... # server-side only, never ship to client
ADMIN_PASSWORD=...               # for the review UI
```

End of document.